



Ricardo Reyes

RESEARCH SCIENTIST · SIMULATION ENGINEER

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Mechanical Engineer with a PhD in Civil Engineering and extensive experience in computational fluid dynamics, finite element methods, and high-performance computing. Experienced across both academic research and industry-facing projects, delivering simulation tools from large-scale HPC environments to applied engineering and infrastructure planning.

Work experience

EAWAG, Swiss Federal Institute of Aquatic Science and Technology

Dübendorf, Switzerland

Scientist, Urban Water Management

Jun. 2024 - Jun. 2025

- Developed computational models (urban water cycle simulation, climate-change scenarios) to support long-term infrastructure planning and policy decisions for the city of Lausanne.
- Built simulation tools for the *UrbanTwin* project enabling rapid scenario testing for environmental impact studies, aligned with the *Swiss Energy Strategy 2050*.
- Performed sensitivity analyses and quality assurance on model outputs, presenting results to external stakeholders including the City of Lausanne and SIE.
- Established reproducible simulation workflows and documentation practices across the team.

EPFL, École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

Postdoctoral Researcher, Chair of Computational Mathematics and Simulation Science

Feb. 2021 - Aug. 2023

- Developed fast surrogate simulation methods (reduced-order modelling) for structural stress, fatigue prediction, and system performance analysis, cutting computation time by orders of magnitude versus full finite element models.
- Implemented and validated numerical algorithms in Python using the FEniCS finite element library.
- Contributed to an *Innosuisse*-funded project with *Akselos*, developing simulation tools for maintenance planning and longevity prediction of offshore wind turbines — directly feeding into their digital twin platform.
- Coordinated with *Akselos* and academic partners, presenting research milestones and supporting integration of simulation tools into industrial deliverables.

Universitat Politècnica de Catalunya

Barcelona, Spain

Doctoral Researcher

Mar. 2014 - Mar. 2020

- Developed VMS-stabilized reduced-order modelling techniques for CFD and structural mechanics as one of the main developers of FEMUSS — an in-house code that has since become a commercial simulation platform.
- Built hyper-reduction algorithms and adaptive mesh refinement strategies to accelerate finite element simulations by orders of magnitude for fluid dynamics and solid mechanics.
- Implemented and optimized numerical solvers in FORTRAN and C++ for incompressible Navier–Stokes, thermal coupling, and fluid-structure interaction problems.
- Ran large-scale simulations on HPC clusters using MPI, PETSc, and HDF5.

Universidad Nacional de Colombia

Bogotá, Colombia

Graduate Researcher

Aug. 2011 - Dec. 2012

- Implemented ANSYS models for industrial problems including fatigue analysis for truck suspension components, turbulence analysis for industrial pump inlet flows, and design optimization of fibreglass lampposts.
- Prepared technical reports, validated models against experimental or field data, and supported engineering teams with simulation-based recommendations.

Education

Universitat Politècnica de Catalunya, BarcelonaTech

Ph.D. in Civil Engineering

Barcelona, Spain

2020

- Thesis: Stabilized reduced order models for low speed flows

Universidad Nacional de Colombia

B.Eng. in Mechanical Engineering

Bogota, Colombia

2010

Skills

Programming	Python, C/C++, Git, Docker, FORTRAN, MPI, LaTeX, Vim
Libraries & Software	HDF5, FEniCS, PETSc, ParaView, Matplotlib, Ansys
Languages	Spanish (Native), English (C2), French (B1, currently studying towards B2)

Publications

JOURNAL PUBLICATIONS

- [1] The Frequency Reduced-Basis Method: Reduced Order Models for Time-Dependent Problems Using the Laplace Transform
Ricardo Reyes
Computational Mechanics, 2026. doi: 10.1007/s00466-026-02795-6
- [2] Reduced order modeling of parametrized pulsatile blood flows: Hematocrit percentage and heart rate
Catalina Farías, Camilo Bayona-Roa, Ernesto Castillo, Roberto C. Cabrales, Ricardo Reyes
International Journal of Engineering Science. 193: p. 103943, 2023. doi: 10.1016/j.ijengsci.2023.103943
- [3] Reduced order modeling for parametrized generalized Newtonian fluid flows
R. Reyes, O. Ruz, C. Bayona-Roa, E. Castillo, A. Tello
Journal of Computational Physics. 484: p. 112086, 2023. doi: 10.1016/j.jcp.2023.112086
- [4] A posteriori error estimates in a finite element VMS-based reduced order model for the incompressible Navier-Stokes equations
Ramon Codina, Ricardo Reyes, Joan Baiges
Mechanics Research Communications. P. 103599, 2021. doi: 10.1016/j.mechrescom.2020.103599
- [5] Element boundary terms in reduced order models for flow problems: Domain decomposition and adaptive coarse mesh hyper-reduction
Ricardo Reyes, Ramon Codina
Computer Methods in Applied Mechanics and Engineering. 368: p. 113159, 2020. doi: 10.1016/j.cma.2020.113159
- [6] Projection-based reduced order models for flow problems: A variational multiscale approach
Ricardo Reyes, Ramon Codina
Computer Methods in Applied Mechanics and Engineering. 363: p. 112844, 2020. doi: 10.1016/j.cma.2020.112844
- [7] Reduced order models for thermally coupled low Mach flows
Ricardo Reyes, Ramon Codina, Joan Baiges, Sergio Idelsohn
Advanced Modeling and Simulation in Engineering Sciences. 5: 28, 2018. doi: 10.1186/s40323-018-0122-7